

The distance between metacentre and _____ is called metacentric height.

- A. water surface
- B. centre of gravity
- C. centre of buoyancy
- D. none of these

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

2. Check valve is used for _____ flow.

- A. very precise control of
- B. unidirectional
- C. multidirectional
- D. none of these

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

3. Capacity of a rotary gear pump can be varied by

- A. changing the speed of rotation.
- B. bleeding air into suction.
- C. bypassing liquid from the suction or discharge line.
- D. all (a), (b) and (c).

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

4. For a given Reynolds number, in a hydraulically smooth pipe, further smoothening _____ the friction factor.

- A. brings about no further reduction of
- B. increases
- C. decreases
- D. none of these

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

5. Power number is the ratio of

- A. drag stress to inertial stress.
- B. inertial stress to drag stress.
- C. inertial stress to gravitational stress.
- D. gravitational stress to drag stress.

Answer: Option A

6. The momentum correction factor for the velocity distribution of laminar flow is

- A. 1.3
- B. 1.66
- C. 2.5
- D. none of these

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

7. For the same flow rate of a fluid, the pressure drop is the least for

- A. venturimeter
- B. orificemeter
- C. flow-nozzle
- D. Δp is same for all

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

8. The valve commonly used in pipes larger than

2" dia is a

A. globe valve

B. plug-cock

C. gate valve

D. check valve

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

9. Foot valves are provided in the suction line of

a centrifugal pump to

A. avoid priming, every time we start the pump.

B. remove the contaminant present in the liquid.

C. minimise the fluctuation in discharge.

D. control the liquid discharge.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

10. An ideal fluid is

- A. frictionless & incompressible.
- B. one, which obeys Newton's law of viscosity.
- C. highly viscous.
- D. none of these.

Answer: Option A

11. In area meter (e.g., rotameter), with increase in the fluid flow rate, the

- A. pressure drop increases linearly.
- B. pressure drop is almost constant.
- C. area through which fluid flows does not vary.
- D. none of these.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

12. Drag co-efficient C_D in Stoke's law range is given by

A. $C_D = \frac{16}{Re,p}$

B. $C_D = \frac{24}{Re,p}$

C. $C_D = \frac{18.4}{Re,p}$

D. $C_D = \frac{0.079}{Re^{0.25},p}$

Answer: Option B

Explanation:



View Answer

Discuss

12. Drag co-efficient C_D , in Stoke's law range is

given by

A.

B.

C.

D.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

View Answer

Discuss

13. In power law, , then the

fluid is

A. Newtonian

B. dilatant

C. thixotropic

D. rheopectic

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

View Answer

Discuss

14. The buoyant force acting on a floating body is dependent on the

- A. viscosity of the liquid.
- B. weight of the liquid displaced.
- C. depth of immersion of the body.
- D. surface tension of the liquid.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

15. A Newtonian fluid is that

- A. which follows Newton's law of motion.
- B. which needs a minimum shear, before it starts deforming.
- C.

for which shear & deformation are related as

- D. none of these.

Answer: Option C

16. In case of end to end connection of two or more pipes in series, the _____ each

pipe.

- A. same rate of flow passes through
- B. head loss is same through
- C. rate of flow in each pipe is proportional to the length of
- D. total flow rate is the sum of flow rate in

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

17. Which of the following equations applies to the fluid flow through a packed bed for very large Reynolds number ?

- A. Fanning equation
- B. Blake-Plummer equation
- C. Hagen-Poiseuille equation
- D. Kozney-Karman equation

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

18. The flow of gas along a pipe in the direction of decreasing pressure causes decrease in its

- A. viscosity
- B. specific volume
- C. velocity
- D. none of these

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

19. Centrifugal pump can't be used to pump

- A. molten sodium (used as a coolant in Fast Breeder Reactor).
- B. moderately viscous vegetable oil used in soap industry.
- C. thick molten soap at 80°C.
- D. none of the above.

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

20. A stream line is

- A. fixed in space in steady flow.
- B. always the path of particle.
- C. drawn normal to the velocity vector at every point.
- D. a line connecting the mid points of flow cross-section.

Answer: Option A

21. Quicksand is an example of a _____ fluid.

- A. bingham plastic
- B. dilatent
- C. Newtonion
- D. pseudoplastic

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

22. Correction for capillary effect in manometers

(used for pressure measurement) need not be applied, if diameter of the manometer tube is _____ mm.

- A. < 4

B. > 4

C. > 12.5

D. < 10

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

23. Differential manometer measures the

A. absolute pressure

B. gauge pressure

C. pressure difference

D. pressure gradient

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

24. The pressure and power requirement of a gas

fan at constant speed & capacity varies

_____ the gas density.

A. directly as

B. inversely as square root of

C. inversely as

D. as square of

Answer: Option A

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

25. Isothermal turbulent flow of a fluid results in decrease of its pressure, which depends on the

A. wall roughness.

B. Reynolds number.

C. both (a) & (b).

D. neither (a) nor (b).

Answer: Option C

26. Boundary layer thickness in laminar flow over a flat plate increases as (where, d = distance from the leading edge.)

A. d

B. $d^{1/3}$

C. d^2

D. $d^{2/3}$

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

27. Gradually varied flow in open channel is a/an _____ flow.

- A. steady uniform
- B. steady non-uniform
- C. unsteady uniform
- D. unsteady non-uniform

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

28. The line of action of the buoyant force acts through the

- A. centroid of the displaced volume of fluid.
- B. centre of gravity of a submerged body.
- C. centroid of the volume of any floating body.
- D. none of these.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

29. Cavitation in a centrifugal pump can be avoided by keeping the

- A. inlet pressure high
- B. outlet pressure low
- C. inlet pressure low
- D. outlet pressure high

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[Workspace](#)

[View Answer](#)

[Discuss](#)

30. Vane anemometer

- A. is an area meter.
- B. is a variable head meter.
- C. rotates an element at a speed determined by the velocity of the fluid

in which the meter is immersed.

D. none of these.

Answer: Option C

31. The peripheral velocity at inlet of a centrifugal pump having inlet diameter of 25 cms and rotating at 950 rpm is _____ m/sec.

A. 1.8

B. 12.4

C. 186.2

D. 736.4

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

32. Laminar flow is characterised by the nonexistence of

A. pressure fluctuation.

B. eddies.

C. deviating velocities.

D. all (a), (b)&(c).

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

33. The length of the tube necessary for the boundary layer to reach the centre of the tube and for fully developed flow to be established is called the _____ length.

- A. equivalent
- B. transition
- C. Prandtl mixing
- D. none of these

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

34. Select the correct practical example of steady non-uniform flow.

- A. Motion of water around a ship in a lake.
- B. Motion of river around bridge piers.
- C. Steadily decreasing flow through a reducing section.

D. Steadily increasing flow through a pipe.

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

35. Hydraulic diameter for non-circular ducts is equal to _____ times the area of flow divided by the perimeter.

A. two

B. three

C. four

D. eight

Answer: Option C

36. Terminal velocity is

A. a constant velocity with no acceleration.

B. a fluctuating velocity.

C. attained after moving one-half of total distance.

D. none of these.

Answer: Option A

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

37. Consider two pipes of same length and diameter through which water is passed at the same velocity. The friction factor for rough pipe is f_1 and that for smooth pipe is f_2 . Pick out the correct statement.

A. $f_1 = f_2$

B. $f_1 < f_2$

C. $f_1 > f_2$

D. data not sufficient to relate f_1 & f_2

Answer: Option C

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

38. For an incompressible fluid, the bulk modulus of elasticity is

A. 5 kg/m^3

B. $\infty \text{ N/m}^2$

C. 1 N

D. 0 N/m

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

39. A fluid is a substance, that

A. has to be kept in a closed container.

B. is almost incompressible.

C. has zero shear stress.

D. flows when even a small shear is applied to it.

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

40. The pressure intensity is the same in all directions at a point in a fluid

A. only when the fluid is frictionless.

B. only when the fluid is at rest having zero velocity.

C. when there is no motion of one fluid layer relative to an adjacent layer.

D. regardless of the motion of one fluid layer relative to an adjacent layer.

Answer: Option C

41. Which of the following must be followed by the flow of a fluid (real or ideal)?

(I) Newton's law of viscosity.

(II) Newton's second law of motion.

(III) The continuity equation.

(IV) Velocity of boundary layer must be zero relative to boundary.

(V) Fluid cannot penetrate a boundary.

A. I, II, III

B. II, III, V

C. I, II, V

D. II, IV, V

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

42. Mach number is defined as the ratio of the local flow velocity to the sonic velocity in the fluid. For what value of Mach number, the

gases are considered incompressible ?

A. < 0.3

B. > 3

C. 50

D. 1

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

43. Boundary layer exists in flow

A. of real fluids.

B. over flat surfaces only.

C. in pipes only.

D. of ideal fluids only.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

44. Rubber latex is an example of _____ fluid.

A. dilatant

B. Newtonian

- C. pseudoplastic
- D. Bingham plastic

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

45. Velocity head on sudden enlargement in a horizontal pipe is converted into _____ head.

- A. elevation
- B. pressure
- C. both (a) & (b)
- D. neither (a) nor (b)

Answer: Option B

46. The frictional resistance in laminar flow does not depend on the

- A. area of surface in contact.
- B. flow velocity.
- C. fluid temperature.
- D. pressure of flow.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

47. A venturimeter measures the

A. velocity head

B. pressure

C. point velocity

D. none of these

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

48. Gradually varying fluid flow is an example of _____ flow.

A. non-steady uniform

B. non-steady non-uniform

C. steady uniform

D. steady non-uniform

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

49. Drag is the force component exerted on an immersed object,

A. passing the centroid of the body at

60° to the direction of motion.

B. the component being parallel to the flow direction.

C. the component being normal to the flow direction.

D. none of these.

Answer: Option B

50. To handle smaller quantity of fluid at higher discharge pressure, use a _____ pump.

A. reciprocating

B. centrifugal

C. volute

D. rotary vacuum

Answer: Option A